Srivatsan Sridharan

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Education

- **Stony Brook University, SUNY** — Stony Brook, NY
  Master of Science in Computer Science — *Aug 2023 – Dec 2024*
    - Distributed Systems, Operating Systems, Databases and Virtualization

Experience

- **Member of Technical Staff 2** — San Jose, CA
  Nutanix — *Jan 2025 - Present*
    - Working on data path services for all I/O to Nutanix's distributed file and block store systems.
    - Introduced a new **Erasure Coding merge operation** to combine multiple short erasure coded strips into wider strips, resulting in **45%** faster garbage collection.
    - Resolved critical system-wide memory leaks by implementing malloc hooks and a leak detection framework, reducing memory usage by **33%** and eliminating a recurring issue that had triggered 24 on-call incidents.
    - Engineered a histogram-based approach to migrate bottom 15% write cold data from SSDs to HDDs across tiers, replacing unbounded heap approaches and reducing peak memory usage of MapReduce jobs by **40%**.
    - Designed and implemented a Quality of Service algorithm, optimizing background task prioritization and batching strategies that improved high-priority task execution by 65% while maintaining overall system throughput of 50K+ tasks/hour across distributed storage nodes.

  Intern — *May 2025 - Aug 2025*
    - Worked on Curator, a MapReduce system handling lifecycle management and background tasks for Nutanix's distributed storage. Implemented job scheduling algorithms, improving cluster health attributes by **20%**.
    - Revamped disk and node rebuild code paths, allowing **15%** faster disk and node rebuild rates.

- **Senior Software Engineer** — Bangalore, India
  Optum (UnitedHealth Group) — *August 2021 - July 2023*
    - Developed highly-scalable distributed systems in Java serving 5M+ users, implementing load balancing algorithms and sharding strategies that achieved **99.95%** availability across multiple data centers.
    - Built components using Java with optimized memory management, reducing system resource usage by **40%** and improving throughput by **55%** on Linux servers.
    - Optimized MySQL database performance by adding composite indexes and implementing connection pooling, reducing query execution time from 2.3s to **180ms** and supporting 3x more concurrent users during peak traffic.
    - Scaled down NLP service calls by deploying a holding tank mechanism, reducing requests by 60%.
    - Implemented Redis caches to decrease microservice chatter by 70% and improving performance by 66%.

Skills

- **Expertise**: Distributed Systems, Operating Systems, Network Programming, Sharding, Consistency
- **Languages**: C++, C, Java, TypeScript, Python
- **Technologies**: Linux/UNIX, Git, SQL, MySQL, PostgreSQL, HTML, CSS, VIM, GCC, Make, TCP/UDP, Algorithms, Data Structures, AWS, Terraform, Azure

Projects

- **RaftKV Store** | *C++, Raft Consensus, Linux*
    - Built distributed key-value store using Raft consensus algorithm with automatic sharding across 8 nodes, implementing leader election and log replication that maintained **99.9%** consistency during network partitions.
    - Designed hash-based sharding strategy with dynamic rebalancing, supporting 10K+ key-value operations/second while ensuring consistency and partition tolerance in CAP theorem trade-offs.
    - Optimized C++ memory management and network I/O using epoll on Linux, achieving **65%** lower latency compared to baseline implementations and handling failover scenarios with **sub-100ms** recovery times.
- **WebCache System** | *C++, Java, SQL, HTML*
    - Built multi-threaded caching server in C++ with LRU eviction policy, serving 500+ concurrent requests with **85%** cache hit rate during stress testing with custom benchmark tools.
    - Designed HTML/JavaScript monitoring dashboard displaying real-time cache statistics, using Git for version control and achieving comprehensive test coverage through unit and integration tests.
- **FileShare Network** | *C++, Python, JavaScript, Linux*
    - Developed peer-to-peer file sharing application using C++ socket programming on Linux, implementing custom protocol that supported file transfers up to 100MB with integrity verification.
    - Created web-based file browser, providing user-friendly interface for file uploads/downloads.
- **SpecFS** | *C, Linux, Algorithms*
    - Implemented speculative file system calls in C, improving file open/close performance by 25% compared to baseline fopen/fclose calls.